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Exercises for Section 1.1: Norm and Inner Product

1. Define the ℓ^1 -norm on n by

$$\|x\|_1 = \sum_{i=1}^n |x^i|,$$

and define the sup-norm on n by

$$\|x\|_\infty = \sup \{|x^i|\}.$$

Show that these satisfy Theorem 1.

Proof.

2. Prove that $\|x\| \leq \sum_{i=1}^n |x^i|$. In other words, the usual norm is no greater than the ℓ^1 -norm.

Proof.

3. Prove that $\|x - y\| \leq \|x\| + \|y\|$. (Compare this with part (2) of Theorem 1.) When does equality hold?

4. Prove that $\left| \|x\| - \|y\| \right| \leq \|x - y\|$.

5. The quantity $\|y - x\|$ is called the distance between x and y . Prove and interpret the “triangle inequality”:

$$\|z - x\| \leq \|z - y\| + \|y - x\|.$$

6. Let f and g be integrable on $[a, b]$.

a) Prove the integral version of the Cauchy-Schwarz inequality:

$$\left| \int_a^b fg \right| \leq \left(\int_a^b f^2 \right)^{1/2} \left(\int_a^b g^2 \right)^{1/2}.$$

Hint: Consider separately the cases $0 = \int_a^b (f - tg)^2$ for some $t \in \mathbb{R}$, and $0 < \int_a^b (f - tg)^2$ for all $t \in \mathbb{R}$.

b) If equality holds, must $f = tg$ for some $t \in \mathbb{R}$? What if f and g are continuous?

c) Show that the Cauchy-Schwarz inequality is a special case of (a).

7. A linear transformation $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is norm preserving if

$$\|T(x)\| = \|x\|,$$

for all $x \in \mathbb{R}^n$, and inner product preserving if

$$\langle Tx, Ty \rangle = \langle x, y \rangle,$$

for all $x, y \in \mathbb{R}^n$.

a) Prove that T is norm preserving if and only if it is inner product preserving.

b) Prove that such a linear transformation is 1-1, and T^{-1} is norm preserving (and inner product preserving).

8. If $T : \mathbb{R}^m \rightarrow \mathbb{R}^n$ is a linear transformation, show that there is a number M such that $\|T(h)\| \leq M\|h\|$ for all $h \in \mathbb{R}^m$. Hint: Estimate $\|T(h)\|$ in terms of $\|h\|$ and the entries in the matrix for T .

9. If $x, y \in \mathbb{R}^n$, and $z, w \in \mathbb{R}^m$, show that $\langle (x, z), (y, w) \rangle = \langle x, y \rangle + \langle z, w \rangle$, and $\|(x, z)\| = \sqrt{\|x\|^2 + \|z\|^2}$. Note that (x, z) and (y, w) denote points in $\mathbb{R}^{n+m}$.

10. If $x, y \in \mathbb{R}^n$, then x and y are called perpendicular (or orthogonal), and we write $x \perp y$, if $\langle x, y \rangle = 0$. If $x \perp y$, prove that $\|x + y\|^2 = \|x\|^2 + \|y\|^2$.

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Exercises for Section 1.2: More Topology:
Open and Closed Sets in $\mathbb{R}^n$

1. Prove that the union of any (even infinite) number of open sets is open. Prove that the intersection of two (and hence of finitely many) open sets is open. Give a counterexample for the intersection of infinitely many open sets.

2. If $A \subset B \subset \mathbb{R}^n$, prove that

$$\text{cl}A \subset \text{cl}B, \quad \text{and} \quad A \subset B.$$

3. Prove that if B is an open subset of A , then $B \subset \text{int}(A)$. Note that this says that $\text{int}(A)$ is the largest open subset of A .

4. Prove that the n -dimensional ball centered at a of radius r ,

$$B^n(a; r) = \{x \in \mathbb{R}^n : \|x - a\| < r\}$$

is open.

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5. Find the interior, exterior, and boundary of the sets:

$$B^n = \{x \in^n: \|x\| \leq 1\},$$

$$S^{n-1} = \{x \in^n: \|x\| = 1\},$$

$$^n = \{x \in^n: \text{each } x^i \text{ is rational}\}.$$

6. If $A \subset [0, 1]$ is the union of open intervals (a_i, b_i) such that each rational number in $(0, 1)$ is contained in some (a_i, b_i) , show that $\partial A = [0, 1] - A$.
7. If A is a closed set that contains every rational number $r \in [0, 1]$, show that $[0, 1] \subset A$.
8. Graph generic open balls in 2 with respect to each of the “non-Euclidean” norms, $\|\cdot\|_1$ and $\|\cdot\|_\infty$. What shapes are they?